MILITARY SPECIFICATION

AMPLIFIERS, AUDIO FREQUENCY; BOOSTER AMPLIFIER - AUDIO - 20 WATT SHIPBOARD ANNOUNCING

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detailed requirements for an audio frequency booster amplifier for shipboard announcing circuits.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-3786	- Switch, Rotary (Circuit Selector, Low Current Capability), General Specification For.
MIL-S-3786/33	- Switch, Rotary, Closed Construction 1-1/2 Inch O.D., 2 Ampere, Style SR.
MIL-R-5757	- Relay, Electrical (For Electronic and Communication Equipment), General Specification For.
MIL-R-5757/15	- Relays, Electrical, Hermetically Sealed, 4 PDT, 5 Amperes.
MIL-L-15098	- Lamps, Glow, General Specification For.
MIL-F-15160	- Fuses: Instrument, Power, and Telephone.
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document shall be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Cont'd)

MIL-F-19207 - Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating, General Specification For.

MIL-A-21577 - Amplifiers, Audio Frequency and Amplifier-Control Groups-Shipboard Announcing.

MIL-E-24142 - Enclosures For Electrical Fittings and Fixtures, General Specification For.

MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types and Associated Accessories, General Specification For.

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring For Electronic Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Booster amplifier. The booster amplifier shall be in accordance with MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirements on this specification sheet shall govern.

3.2 General requirements. The booster amplifier shall conform to the schematic and wiring diagrams as specified (see figures 1 and 2).

3.3 Switch. A switch shall be provided. The switch shall be in accordance with MIL-S-3786 and MIL-S-3786/33, D construction, type SROOD22B1MPO.

3.4 Relay. The K relay shall be R44 hermetically sealed plug in type in accordance with MIL-R-5757 and MIL-R-5757/15 (see figures 1 and 2). There shall be four double throw isolated poles. The relay coil operating voltage shall be 24-32 V d.c. This voltage shall be supplied from the amplifier d.c. supply so that failure of this supply will deenergize the relay, and pass the audio signal unboosted on to the speaker load.

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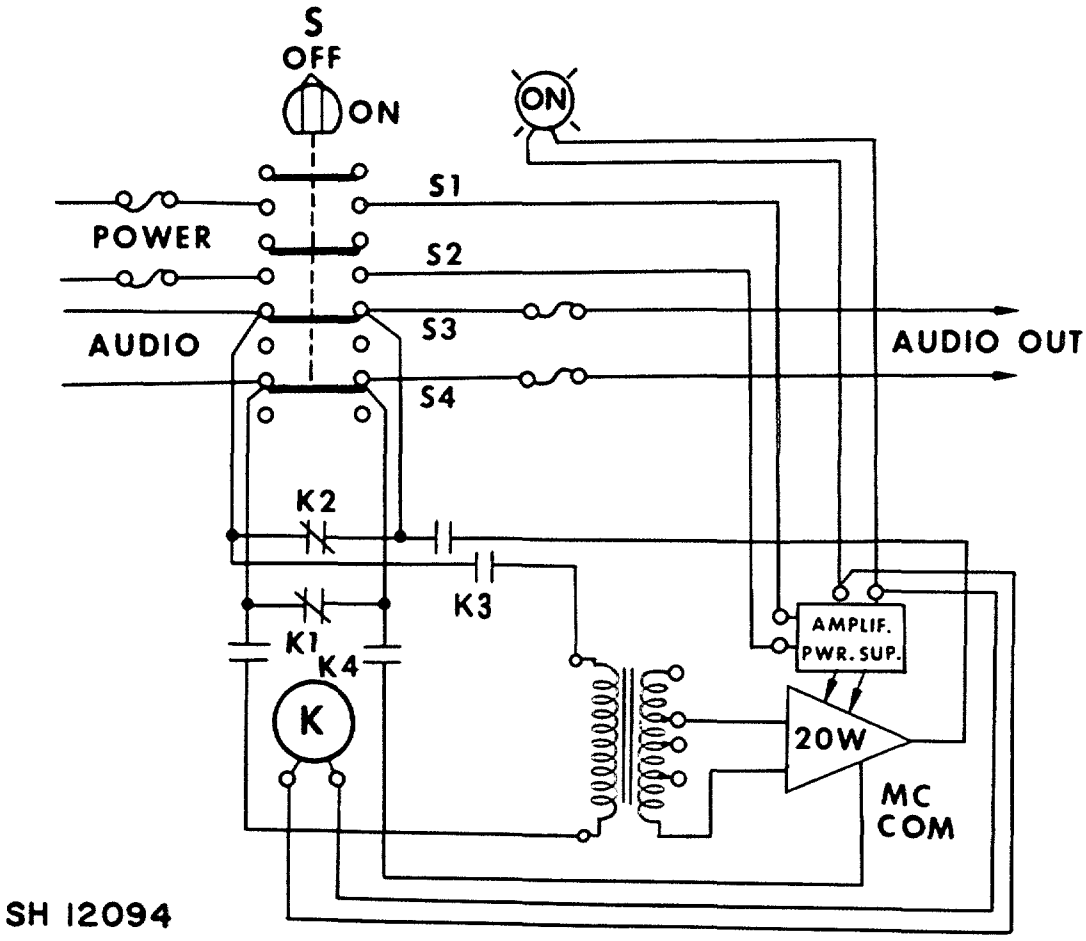


FIGURE 1. Booster amplifier schematic diagram.

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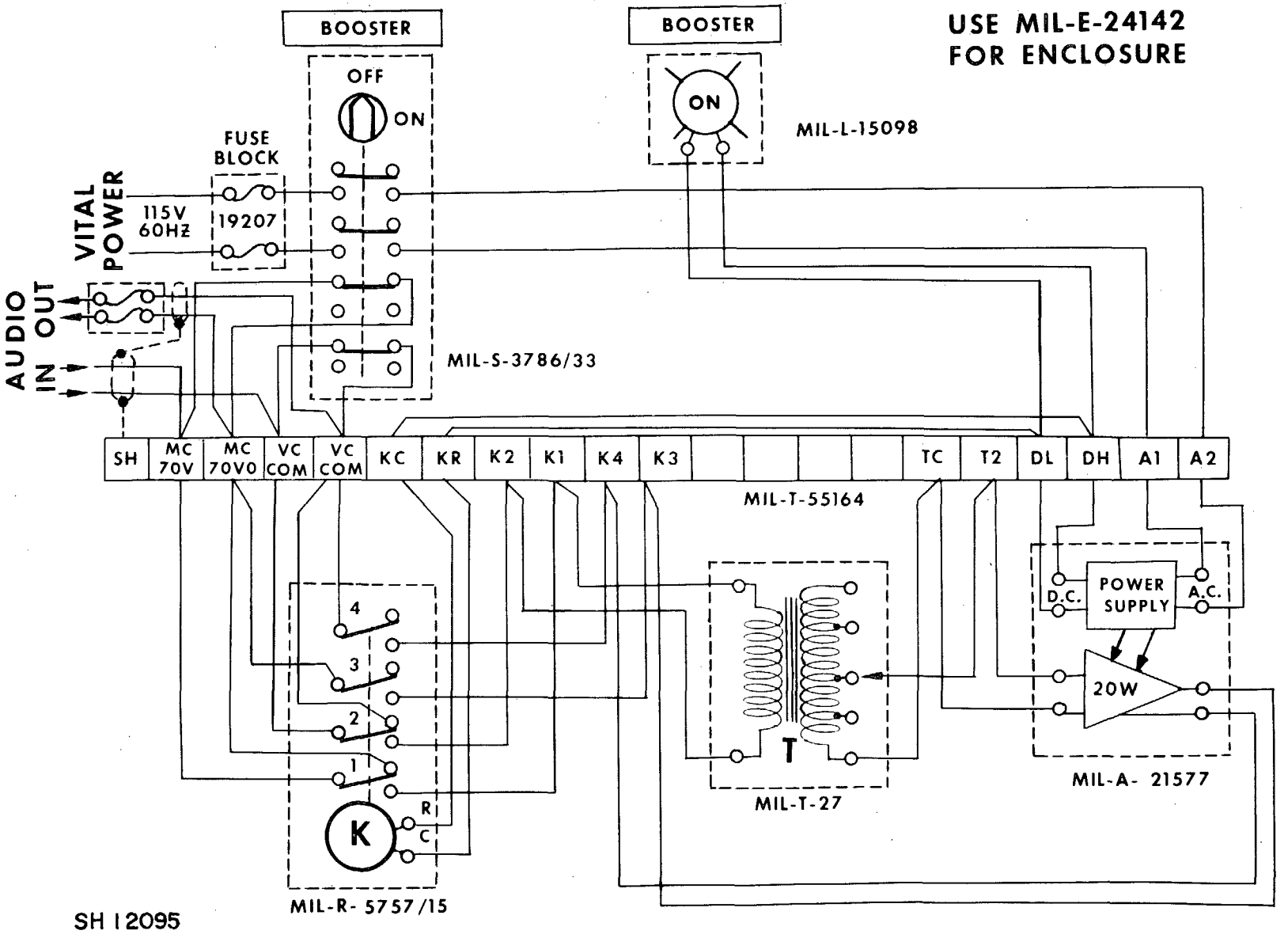


FIGURE 2. Booster amplifier wiring diagram.

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3.5 Transformer. The audio input transformer shall be in accordance with MIL-T-27 audio transformer, as required for use in a 70 volt audio system.

3.6 Amplifier. If the contractor has produced and supplied to the Navy a 20 watt amplifier acceptable within the requirements as specified herein, the same electrical components shall be used. The assembly may be rearranged as necessary for the selected enclosure (see 3.9).

3.7 Lamp. The "ON" lamp shall be a glow lamp conforming to MIL-L-15098. The lamp shall indicate when the amplifier d.c. supply voltage is available, either directly or by using relay K contacts.

3.8 Terminal board. Terminal boards shall be in accordance with MIL-T-55164.

3.9 Enclosure. The booster amplifier compliment shall be enclosed in accordance with MIL-E-24142. Painting requirements shall be in accordance with MIL-E-16400 for interior use. Size shall not exceed 110 cubic inches. The cover plate(s) shall have retained screws with slotted hex heads and shall be hinged or chained (with sleeving) to the mounted portion of the enclosure.

3.10 Fuses. Fuses shall protect the amplifier power supply and the amplifier. Fuses and fuseholders shall be selected from MIL-F-15160 fuses and MIL-F-19207 fuseholders as specified in MIL-STD-1360.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling, examinations and tests shall be as specified in MIL-A-21577.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in MIL-A-21577.

6. NOTES

6.1 The notes specified in MIL-A-21577 are applicable to this specification.

Preparing activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-21577/11(SH)

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